

(a) Contrastive extraction

same unnamed stem · 30 pairs / domain

expert (+)

You are a 3106 practitioner. What is a culture?

default (-)

You are a helpful assistant. What is a culture?

residual stream



mean-pool assistant h



$$v = \text{unit}(\mu+ - \mu-)$$

144 ANZSRC domains · layers 10–20 · exam terms held out of v

(b) Residual-stream add

$$h \leftarrow h + \alpha \|h\| v \quad \text{one decoder block}$$



$\alpha \|h\| v$

added at one layer
norm-relative shift
every new token

$$\alpha = 0.10 \dots 0.40$$

report the full

11 × 7 grid

Llama-3.1-8B · layers 10–20

(c) Held-out polysemy exam

15 traps × 144 domains = 2,160 items

Q What can a fingerprint distinguish?

gold product identity (3106) · not a person's ID

v off

v on

unsteered

helper prompt

CAA

helper + v

prompt

names the domain

prompt+CAA

name + v

score: MPNet cosine to one frozen medoid



α (0.10 ... 0.40)